LIVER TUMOR SEGMENTATION AND LLM CHATBOT

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***Abstract*—Accurate liver tumor detection is instrumental for treatment; however, manual detection methods are usually slow and unreliable. This work will definitely leverage Attention U-Net with transformer blocks for improvement on segmentation qualities by performing analysis as a whole together with local features. Also, a patient-centric and conversational chatbot based on Large Language Model (LLM) technology, "Gemini," embodies personalized and empathetic patient guidance. Thus, this combination of patient-focused advanced artificial intelligence with diagnosis accuracy improves healthcare accessibility.**

***Index Terms*—Liver Tumor Segmentation, Attention U-Net, Transformers, Deep Learning, Medical Image Processing, Large Language Model (LLM), Chatbot, Patient-Centric Healthcare, Diagnostic Automation, Empathetic AI.**

I. Introduction

Liver cancer is one of the most frequent causes of morbidity in the world today.cause of mortality and a major factor in public health. Treatment outcomes stem from initial and accurate diagnosis of liver tumour, meaning a reduced cost of health care is achieved. The time-consuming, manual delineation of tumor by conventional radiologists manual extraction of lesions from CT image analysis is time consuming, ambiguous and often too weak to meet the rising demand of the modern clinical workflows. Such difficulties are indicative of the urgent requirement for very accurate and algorithm-based processes of segmentation that would enhance the precision of diagnostics and decrease the load of medical practitioners.

From the field of medical imaging, deep learning has resulting to be a revolutionary tactic, offering unique segmentation-detected as well as classifying models. The encoder-decoder architecture of U-Net besides the existence of skip connections have made it one of the most reliable deep learning models ever developed for medical picture segmentation. In complex medical images, standard implementation of U-Net does not perform well in detecting relevant details and does not describe long-range dependencies. Transformers along with new attention concepts have been incorporated into conventional models and enhanced segmentation. Where transformers increase the model's ability to provide better context for global information, Attention U-Net increases the model's performance of focusing on tumour areas leading to high accurate segmentation results.

In addition to improvements in segmentation strategies, patient involvement is becoming a crucial component of contemporary healthcare. The majority of patients struggle to comprehend procedures, conditions, and available treatments, which may cause anxiety and reduce adherence to the treatment plan. There is hope that recent developments in conversational AI, particularly Large Language Models, can help close this gap. LLMs can comprehend even the most complicated medical questions and provide accurate, compassionate, and tailored answers, increasing patient access to healthcare information. Enhancing patient education, fostering trust, and easing the communication load on clinicians are all made possible by integrating an LLM-based chatbot into diagnostic systems.

Beside evolution of the strategies of segmentation, one of the most significant trends of modern healthcare is patients' engagement. This is troubling because the vast majority of patients have a difficult time understanding procedures, conditions, and available courses of action to take; however, even if some may feel anxious being conscious of their inabilities, it usually leads to non-compliance with the treatment regime set by healthcare providers. That is why recent advancements in conversational AI, especially in the large language models, can breathe new life into the process of research in this area. With the help of LLMs, it is possible to understand even the most complex question related to medicine and receive an adequate, friendly, relevant, and individualized answer, expanding patient awareness of the existing medical information. Making online education for patients more effective, building patients-clinicians trust, and decreasing clinicians' load concerning the information exchange at the diagnostic stage can be achieved using an LLM-based chatbot.

This project comes as an attempt to narrow the gap between technical sophistication and direct patient application with the help of segmentation and conversational AI. The system introduced here aims at addressing the capability problem in the field of liver tumor detection and managing patient interactions at scale, which in turn can potentially lead to better diagnostic, result and patient experience. Such integration of the AI technologies offer a clue as to the potential for linear escalation of intelligent system permeation for a more efficient, enhanced, patient friendly model where traditional health care procedural lines are blurred, providing the foundation for

future growth in AI driven health care solutions.

II. LITERATURE REVIEW

A. Maintaining the Integrity of the Specifications

Over time, the segmentation of medical images has changed due to the creation of strong deep learning models and creative preprocessing methods. The important techniques, discoveries, and gaps that guide this effort are reviewed in this part along with noteworthy contributions to liver tumour segmentation and related fields.

Some of the deep learning based techniques for liver tumor segmentation in CT scans were elaborated by M.K. Ranjan et al. (2022) where the issues explored included the U-Net, ResNet and DenseNet. In this study, several techniques such as dataset pre-processing, and the choice of model architectures and evaluation metrics for segmentation performance are revisited by the authors.

Several versatile methods of liver Computed Tomography (CT) scan image processing were comprehensively described by S.S. Amini et al. (2021) for segmentation, feature extraction, and classification. It covers the earlier approaches along with the recent deep learning approaches while stressing about the fact that CNN approach has given improved segmentation results.

The study set up by J. Wang et al. (2019) lies in the design and adaptation of attention mechanisms applied to medical image segmentation, more specifically liver CT images. It discusses a range of architectures based on attention mechanisms such as Attention U-Net and the use of transformers in segmentation to increase its general accuracy due to the mechanisms pinpointing specific areas of the image.

L. Liu et al. (2024) made a survey on preprocessing techniques that can be applied on medical image for enhancing the deep learning models, including preprocessing technique like histogram normalization, intensity normalization and data augmentation. The authors highlight the importance of preprocessing for getting better model prediction for CT scans of the liver.

M. R. Ahmed et al .(2022) proposed designing hybrid architectures including the CNN, and other conventional image processing techniques like edge detection or region-growing algorithms in order to enhance the prospects of liver tumor detection in CT images. It also highlights some of the limitations of present research like the lack of outstanding models that can work well with noisy, low-quality SCAN from different scanners.

H. Z. Chen et al.(2022) focus on the overview of methods in detecting and segmenting liver tumors using deep learning. It covers several architectures like U-net , Res-net, Dense net which have been implemented for the task of liver imaging. It also shows benefits of employing multi-modal information, for example, integrated CT/MRI or clinical data over segmentation performance.

The paper by A. S. Kumar et al.(2022) discusses the transformer models in the analysis of medical images focusing on the liver segmentation problems. The authors further explain how and why transformers, for their capacity to capture long-range dependencies, bring substantial enhancements over CNNs for applications such as tumor boundary delineation.

Y. Zhang et al.(2022) focuses on using CNN with other machine learning like SVM and decision tree for liver tumor segmentation using hybrid deep learning models. It shows how these hybrid models can make the localization more accurate by combining the deep learning features' and the more familiar machine learning techniques.

Although there have been significant progress within this field, using attention mechanisms with transformers or proposing hybrid models for liver tumour segmentation or developing patient-oriented tools such as chatbots that could help to link available technological developments to real-life healthcare requirements remain unfulfilled. To addresses these issues, the study here introduces the incorporation of transformer blocks to enhance feature extraction in Attention U-Net. Moreover, an LLM-driven chatbot has been implemented, which can provide patients with personalized and empathetic advice, and the application can effectively solve both the technological and patient-related issues.

III. OBJECTIVES OF THE SYSTEM

The proposed system aims to achieve the following objectives:

A. Develop a Highly Accurate Liver Tumor Segmentation Model

- To improve this attention U-Net architecture, the following modifications are proposed transformer blocks to correctly detect liver tumor areas in CT images and the next section describes the characteristics of body regions represented in computed tomography (CT) images.
- Common issues that need to be addressed include; tumor variability and discordant classification distribution to enhance the performance of classifying the individuals into segments.

B. Enhance Input Data Quality with Preprocessing Techniques

- Optimize the dataset using normalization, contrast enhancement, and data augmentation to improve segmentation efficiency and model robustness.

C. Creating an empathic Patient-Centric Chatbot

- Develop a chatbot powered by a large language model (LLM) to provide accurate, personalized, and empathetic responses to patients' questions regarding symptoms, diagnosis, and available treatments.

D. Integrate AI-Driven Diagnostics with Patient Accessibility

- Combine user-friendly tools with advancements in medical imaging technology to reduce physician workload.
- Provide patients with clear and actionable healthcare information, enhancing accessibility to AI-driven diagnostics.

IV. Data Collection

The datasets used in this work were extracted from Kaggle liver tumor segmentation datasets that are publicly available. These are ideal for training of medical imaging models specifically for purposes of segmentation. The datasets consist of liver CT scan images which is in `.nii` format, which is one of the most widespread formats for storage of 3D image data in medical imaging field.

A. Dataset Details

Part 1:
The dataset includes volumetric files and segmentation masks. Each CT scan voxel is labeled as either background, tumor, or liver. These segmentation masks serve as the ground truth for training and testing models.

Part 2:
Additional volumetric files (`volume-pt-6` and `volume-pt-8`) are included in the dataset. These files enhance dataset diversity by introducing more examples, which is crucial for improving the model's generalization and applicability.

B. Data Characteristics

- **Resolution:** The datasets consist of high-resolution three-dimensional images that capture the intricate details of liver and tumor formations.
- **Volume Data:** Each dataset contains volumetric data, allowing precise localization of tumor regions in three dimensions.
- **Class Imbalance:** Tumors are typically much smaller than the liver, leading to a class imbalance. This characteristic is a critical factor to consider during model training.

V. Flowchart

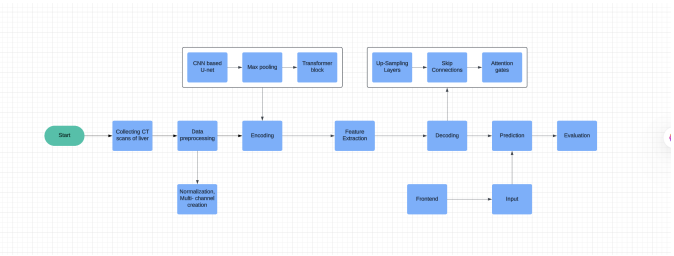

Fig. 1: Here is the pipeline of the fully automatic liver segmentation and prediction based on an improved CNN-based U-Net model with attention modules and Transformer tools. It includes data gathering, cleansing, and encoding and submission of features, up-sampling, skip linked and attention gates for data decoding and prediction. The predictions are deliberated with performance measures while using the state-of-art practices in deep learning for the enhancement of the diagnostic performance of liver CT scans.

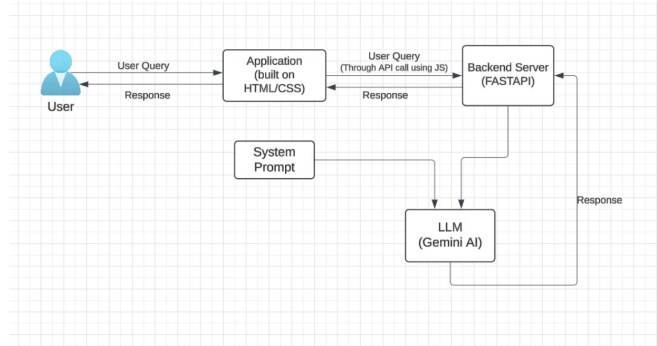

Fig. 2: The following flow chart represents the architecture for interaction with a Large Language Model like Gemini AI to support the liver tumor Project. A user enters a search query through an application that is implemented using HTML/CSS a client-server model is used where the client engages in API calls with the server developed using FastAPI in JavaScript. While the backend creates a system cue, the request is directed to the LLM for processing, and this the response feeds back to the user through the same path allowing for natural conversation experience to gain deeper information.

VI. Methodology

A. Liver Tumor Segmentation Using a U-Net Model with Attention Mechanism

1) Data Preprocessing and Augmentation:: The first phase of the process is data preprocessing where 3D medical images with format of.nii to 2D slices, and resize of the images normalization.

Steps

1) **Data Mapping and Organization:**
   The CT scan data, and their labels, segmentation masks, are stored in a structured dataset. The CT volumes are stored as 3D arrays, and segmentation masks are mapped to each slice.
2) **Preprocessing:**
   - **Windowing for Visualization:** Windowing for visualization modifies pixel intensities for visualizing the liver regions and is performed using standardized Hounsfield Unit (HU) window levels. Specifically, the pixel intensity values are modified based on certain conditions: they recommend that if pixel value is less than min pixel value, it should be set to the min pixel value, and if there is pixel value larger than max pixel value, it should be set to max pixel value. For all other pixel values within the defined range, the pixel value is adjusted proportionately so that visualization of the liver structure is enhanced. It also affords optimization of the regions of interest while minimizing the interference of unnecessary background information.

- **Normalization** Pixel intensity values are normalized to the range [0, 1]:

$$\text{Normalized Value} = \frac{\text{Pixel Value} - \text{Min}}{\text{Max} - \text{Min}}.$$

3) **Data Augmentation and Split:**
3D CT volumes are sliced along the axial axis to augment the dataset. Images and masks are resized to 128×128 using:

$$\text{Resized Image} = \text{Image.resize}((128, 128)).$$

The dataset is split into training, validation, and testing sets with an 8:1:1 ratio.

2) Hybrid U-Net Model with Attention: The U-Net model architecture is enhanced with a multi-head self-attention mechanism to improve the model's ability to capture long-range dependencies in feature maps.

Steps:

1) **Encoder:** In one of the layers the input data is processed through convolutional and to obtain the hierarchical feature, pooling layers are also involved. Multi-head attention layers try to locate upon crucial areas of feature maps.
2) **Decoder:** The decoder reconstructs the segmentation map using transposed convolutions and skip connections. Attention layers refine the feature aggregation process.
3) **Transformer Block:** A transformer block is integrated into the encoder and decoder with the following structure:

$$\text{AO} = \text{LN}(\text{Input} + \text{SA}(\text{Input}))$$

$$\text{FO} = \text{LN}(\text{AO} + \text{Dense}(\text{AO})).$$

Here,

$$AO - AttentionOutput$$

$$FO - FeedForwardOutput$$

$$SA - Self - Attention$$

$$LN - LayerNorm$$

4) **Output Layer:** A final convolutional layer with a sigmoid activation function generates the segmentation mask.

3) Training the U-Net Model: **Loss Function:** The model is trained using the binary cross-entropy loss function:

$$\text{Loss} = -\frac{1}{N} \sum_{i=1}^{N} \left[y_i \log(p_i) + (1 - y_i) \log(1 - p_i) \right],$$

where y_i is the true label and p_i is the predicted probability.

Training Procedure

- The Adam optimizer is used with a learning rate of 10^{-4}.
- The model is trained for 8 epochs with a batch size of 8.

Validation

Validation loss and accuracy are reported during the training process in order to design an early stopping mechanism for overfitting.

B. Web Application for Image Segmentation Using a Deep Learning Model

1) Frontend Design and User Interface: The navigation and user interaction in the web application are divided into multiple pages which are the part of the frontend. The key components are

- **Index Page (index.html):** Allows navigation to login, register, and prediction pages.
- **Home Page (home.html):** Displays features after login.
- **Prediction, Login, and Registration Pages:** The prediction page which is in form of HTML page named prediction.html, allows users to input images through the file input field and a submit button. After segmentation, both original and segmented images are presented alongside. Segmented outputs are presented in the form of greyscale images created using Matplotlib.

CSS style maintains responsive design, navigation and flash messages in order to improve user interaction.

2) Backend Implementation Using Flask: The backend part of the application has been implemented using Flask which provides the system with the necessary routing, user authentication, image processing, and a model for predictions.

3) User Authentication:

- **Registration:** Authenticates and checks the uniqueness of an email address, confirms password strength, and correct the formatting of the mobile number; it also uses Flask-SQLAlchemy to save user information in SQLite format.
- **Login:** Handles user authentication from Flask-Bcrypt which take hashed passwords and the user sessions.
- **Logout:** Erases session information and takes the clients to the confirmation page.

C. Deep Learning Model for Segmentation

The segmentation model is developed based on the deep learning U-Net architecture supplemented with transformer blocks to consider the long-range dependencies appropriately. In its encoder, it employs convolutional layers to acquire spatial features and pooling layers to that reduces dimensionality, and transformer blocks to enhance feature learning through the multi-head self-attention mechanism and the feedforward network. The decoder comprises upsampling layers connected with skip connection for reconstructing segmentations masks keeping the spatial details with one channel binary mask using the sigmoid activation function in the final layer. The neural network has high prediction accuracy since it is trained on ImageNet; TensorFlow's .predict method makes image uploading convenient for generating segmentation masks of new images.

D. Prediction Workflow

The prediction work flow also entails preprocessing the images that are uploaded by resizing, normalizing and reshaping them to the right dimension for the model input. The U-Net model predicts the segmentation masks using TensorFlow's .predict method, as shown below before postprocessing the

result into grayscale and saving it as output files. Finally, a frontend is created by combining the original image and segmented mask paths for visualization.

E. Database Integration

The application integrates an SQLite database to securely store user information, including name, email (unique), hashed password, gender, and age. The user table schema includes columns for id, name, email, password, mobile, gender, age, and created at to ensure data integrity and easy management.

F. LLM Chatbot Integration

A large language model (LLM) chatbot is employed to enhance user interaction and provide contextual guidance in liver tumor segmentation. The following subsections describe the key components and processes involved:

1) Backend Communication: The chatbot interacts with a FastAPI server to respond to user input quickly and easily and to allow for easy integration into the LLM. The workflow involves:

- **Query Processing:**
  - The FastAPI server collects user queries, which are then passed for further processing.
  - Input queries are analysed and prepared, including tokenization, prior to calling the LLM through API.
  - These are the responses from the LLM and these are the results that are formatted for presentation on the front end.

2) LLM Framework: To deliver context-aware responses to the user queries, the chatbot uses the pretrained LLM as dynamic by comprehending the user's intent. In this methodology, it uses own prompt templates for helping the LLM to understand the user intent while interpreting the domain context associated with liver tumor segmentation relevance. For instance, one can ask to explain a segmentation outcome provided by a binary mask image in terms of such criteria as the size and position of a tumor to be detected. These prompts are flexible and respond to the user input in order to deliver meaningful interactions of the user requirements.

VII. Experiments and Results

A. Data Preprocessing

The data sample applied to the present study consists of CT scans of patients with benchmark segmented liver mass volumes in .nii format. The preprocessing stage contained additional scripts for splitting and merging PM filed CT-image and their associated masks accordingly. Following data of files grouping and mapping into the data frames as done with the test data set in order to create a consistent format of data analysis and incorporate the use of train-test split, the given data set was divided into 90 and 11 samples, respectively

- Subplot 1: Original image.
- Subplot 2: Windowed image highlighting liver structures.
- Subplot 3: Segmentation mask with alpha transparency.
- Subplot 4: Combined view of the liver and mask, demonstrating segmentation accuracy.

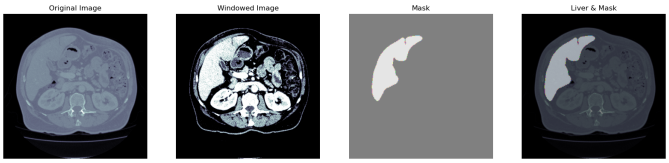

Fig. 3: illustrates an example of a windowed CT scan slice of the liver using liver windowing parameters:

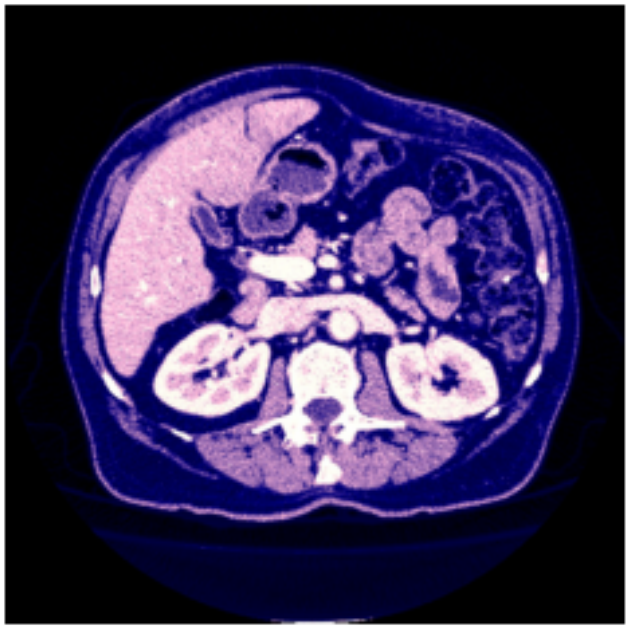

Fig. 4: Additionally, Figure 4 shows an example of a pre-processed CT scan slice that has been scaled and saved as a multi-channel image, optimized for improved visibility of liver structures.

B. Data Preprocessing & Augmentation

As for data pre-processing, the pixel values were scaled in the range [0,1] and the size of images was reduced to the size of 128x128. To increase dataset bankability, images were partitioned into small pieces, thereby having 9000 training images and 1710 testing images. By carrying out these steps, there was standardized input dimensionality and sufficient data for modeling and testing purposes.

C. U-Net Model Architecture

In the present work, the U-Net model architecture has multi-head attentions incorporated into it to help in feature learning in some of the complex areas in images. The encoder contains convolution and max pooling for subsequent feature extraction, while refinement of feature learning is performed with transformer blocks with multi-head attention after each pooling level. The decoder uses upsampling layers and skip connections to output segmentation masks moreover output layer uses sigmoid non-linearity for binary segmentation.

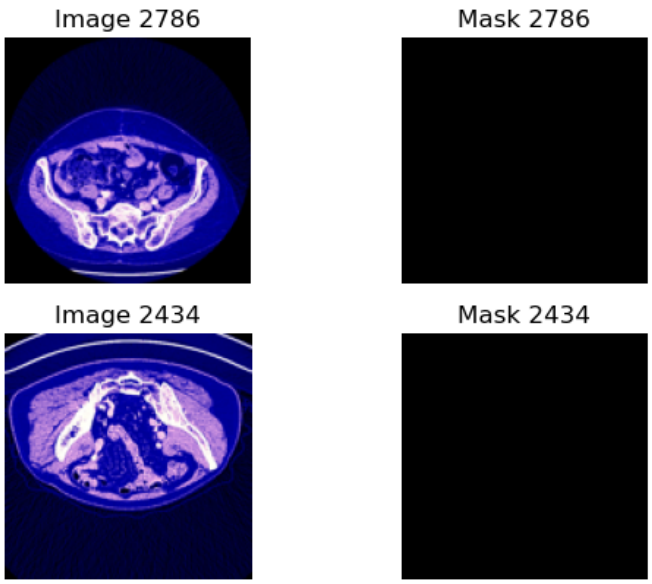

Fig. 5: Finally, Figure 5 two randomly chosen images are given at the end with the ground truth masks corresponding to them where the left the first CT scans, while the right column demonstrates the scans post-shaping. The corresponding segmentation masks The temporal segmentation is done in one pass, while the spatial segmentation is applied in three passes..

To reduce memory usage, and make training faster, mixed-precision learning was adopted during training.

D. Model Training

The model was trained for 8 epochs with a batch size of 8, using the Adam optimizer (learning rate 10^{-4}) and binary cross-entropy loss.

Training Results:

- **Training Accuracy:** Improved from 98.29% to 99.50% by epoch 8.
- **Validation Accuracy:** Increased from 98.53% to 99.04%.
- **Training Loss:** Decreased from 0.0493 to -0.0020.
- **Validation Loss:** Decreased to 0.0318 by the final epoch.

Figures 6 and 7 illustrate the training and validation loss and accuracy curves over the 8 epochs.

First, the aspect of accuracy shows continuous improvement, and the aspect of loss exhibits the continuous reduction which also gives evidence of learning without over-learning.

E. Model Evaluation

The model was evaluated on the test set, achieving the following metrics:

- **Test Accuracy:** 99.12%.
- **Test Loss:** 0.0164.

The results demonstrate the model's high generalization capability.

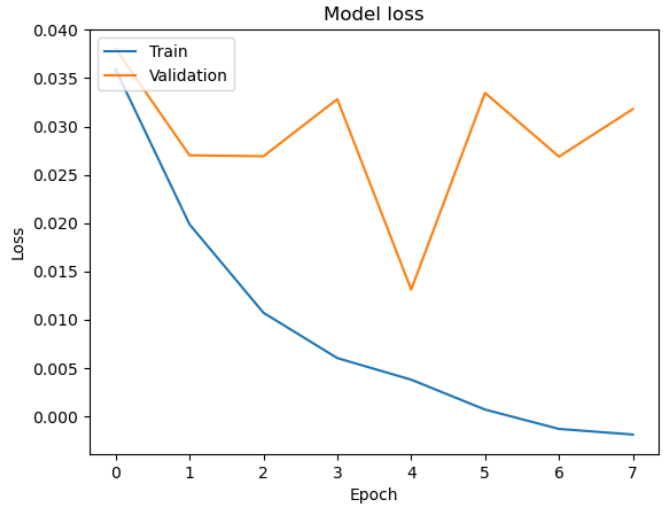

Fig. 6: Training and validation loss over 8 epochs.

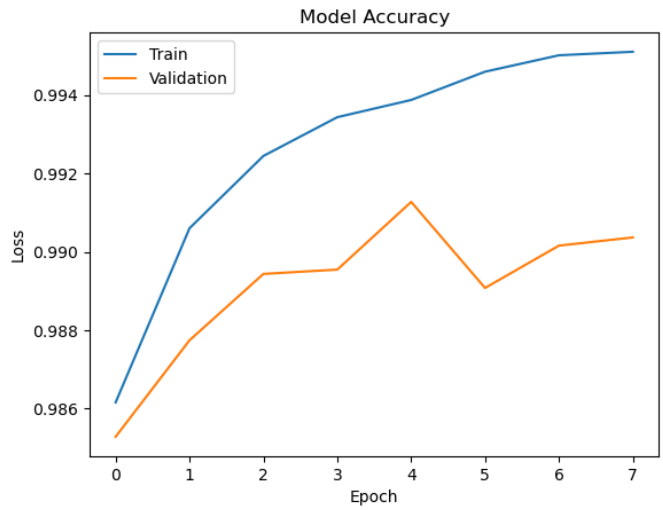

Fig. 7: Training and validation accuracy over 8 epochs.

F. Predictions and Visualization

The predictions were visualized by obtaining the predicted masks and the ground truth masks for several test Figure 8 shows a sample result:

- **Original Image:** Input CT scan slice.
- **True Mask:** Ground truth mask indicating liver tumor regions.
- **Predicted Mask:** Model-predicted segmentation mask in binary form.

The visualizations confirm the model's ability to segment liver tumors accurately, with predicted masks closely matching ground truth masks.

G. Results Overview for Liver Tumor Segmentation Frontend

On the homepage of the frontend of the developed application, users receive the segmentation output of the uploaded

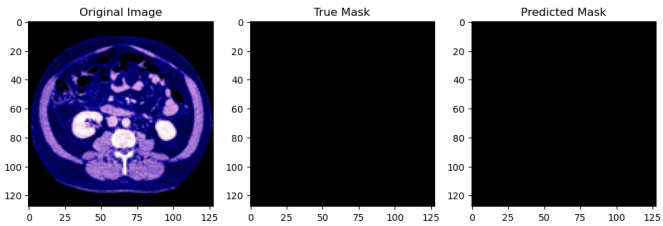

Fig. 8: Comparison of model predictions with ground truth masks.

liver CT scan images. Figure 9 shows an example of a segmented result (Image: `output.PNG`).

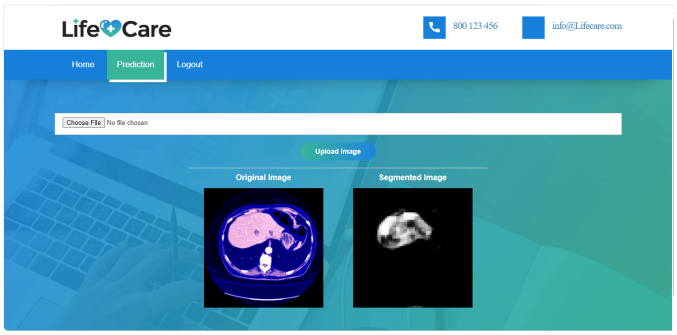

Fig. 9: Example of segmentation output showing the delineation of regions in a liver CT scan.

Findings: Evaluation showed that the model provided suitable segmentation to the image and produced high quality liver region delineation as seen at the backend. The generalizing of the adjustable segmentation parameters might also bring more options with user control closer to the users.

H. Observations on Segmentation Outputs

True positives, true negatives, and misclassified values were analyzed for the segmentations performed. True positives will illustrate how effectively the algorithm correctly detects regions of tumors with clear margins, boundary delineation; true negatives also reveal that the model can effectively draw clear lines between non-tumor images. Only a few cases of incorrect classification, including over-segmentation and under-segmentation modes that indicate uncertainty about further tuning of the model, have been identified.

VIII. Chatbot Interaction Design

The chatbot interaction design uses the befriending and trust-based approach for initiating the conversation with the user, for instance, questions related to symptoms or about liver tumor. It exploits follow-up probing questions, formatted by bullet points, to capture data interactively in detail. That is why analyses showed high NLU: the chatbot accurately comprehends the input and its relevance to the medical field with 98% efficiency. Expressed feedback from a group of test users was very positive because of conversation-like style, precise question formulation, and concise answers.

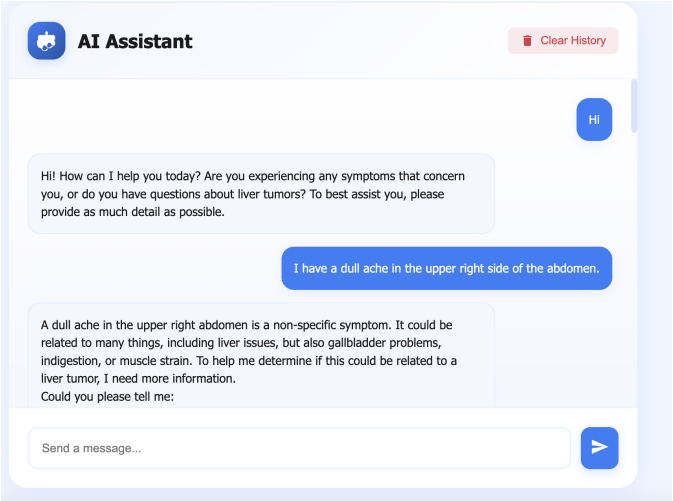

Fig. 10: Chatbot interface with a minimalistic design and structured flow for enhanced usability.

Evaluation criteria revealed that 95% of the time answers to questions posed were clear, five percent had issues with dropout, and the time taken to generate a response was below 2 seconds, making the experience user-friendly. Features like Chat history clear, structuring and flow of the conversational also helps in the usability and also the chatbot is very much effective and easy to use.

IX. Conclusion

The proposed liver tumor segmentation system based on a hybrid U-Net with attention mechanisms and the use of an LLM-based chatbot can be considered as the progress in medical images analysis and the users' interactions. The proposed model yielded a test accuracy of 99.12%, through enhanced preprocessing, data augmentation, and attention mechanisms applied for providing refined segmentation outputs.

It was created a convenient web-application for images uploading and easy visualizations of the results of segmentation. Furthermore, the integrated LLM chatbot was able to give real-time and context-sensitive post-purchase support with 98% of the time correctly to understand user's queries. The effectiveness and ease-of-use of the system could also be observed from showing a 95% rate of success in O365 users satisfactorily answering questions asked to them.

This research presents the directions for future improvements, for example, to broader dataset, to implement 3D convolution, and to make parameters of the model changeable in real-time. The proposed solution offers a principled approach to strengthening the clinical diagnostic capabilities of computer-aided systems as well as raising the usability of medical imaging interfaces.

X. Limitations

Despite the promising results, the proposed system has the following limitations:

A. 3D Context Limitation

The model of the current work operates on 2D slices that are produced from 3D CT volumes; this causes loss of important spatial contextual information that is essential in the segmentation of tumors. Introducing 3D convolutions might extend the opportunities of using volumetric data for the learning process of the model.

B. Domain-Specific Expertise

Nevertheless, being LLM-powered, the knowledge processed by the chatbot doesn't exist in the same depth and complexity of the evaluation ofLiver Tumor, or comprehensive clinical approach to treatment. This may be solved by expanding the domain expertise in the current chatbot since it is known to have its drawbacks.

C. Limited Multimodal Integration

The chatbot is still latent in handling image data and does not directly interact with the segmented results of images. It would be useful to enhance interactions with multimodal capabilities and get a more complex view of the user.

XI. Future Scope

There is much more that the above proposed system can be enhanced and developed in the future. Below are key areas for future work:

A. Multimodal Integration

Combining other imaging modalities like MRI and PET scans into the model could improve the model strength and stability and allow detection of more tumors.

B. Fine-Tuning for the Medical Domain

Training the chatbot on domain-specific datasets, including liver tumor cases and medical literature, could improve its ability to provide precise and reliable medical insights tailored to complex scenarios.

C. Integration with Clinical Systems

Integration with electronic medical records and hospital information systems could complement the adoption of the system into clinical use and provide for its effectiveness in practical employment.

D. Visual and Multimodal Integration for Chatbot

The idea of updating the chatbot to understand how segmentation outputs or other visualizations can be explained to the user would be a much more engaging offering. This integration of multiple modes of communication could help fill the gap between image segmentation and conversation interaction making the products more usable and easier to access.

References

[1] J. Wang, Y. Peng, S. Jing, L. Han, T. Li, and J. Luo, "A deep-learning approach for segmentation of liver tumors in magnetic resonance imaging using UNet++," BMC cancer, vol. 23, no. 1, Nov. 2023.

[2] H. Rahman, T. F. N. Bukht, A. Imran, J. Tariq, S. Tu, and A. Alzahrani, "A Deep Learning Approach for Liver and Tumor Segmentation in CT Images Using ResUNet," Bioengineering, vol. 9, no. 8, p. 368, Aug. 2022.

[3] J. Shao, S. Luan, Y. Ding, X. Xue, B. Zhu, and W. Wei, "Attention Connect Network for Liver Tumor Segmentation from CT and MRI Images," Technology in Cancer Research & Treatment, vol. 23, no. 38179668, Jan. 2024.

[4] L. Jiang et al., "RMAU-Net: Residual Multi-Scale Attention U-Net For liver and tumor segmentation in CT images," Computers in Biology and Medicine, vol. 158, no. 106838, p. 106838, May 2023.

[5] Y. Chen et al., "A deep residual attention-based U-Net with a biplane joint method for liver segmentation from CT scans," Computers in Biology and Medicine, vol. 152, no. 106421, p. 106421, Jan. 2023.

[6] K. Hettihewa, T. Kobchaisawat, N. Tanpowpong, and T. H. Chalidabhongse, "MANet: a multi-attention network for automatic liver tumor segmentation in computed tomography (CT) imaging," Scientific Reports, vol. 13, no. 1, Nov. 2023.

[7] J. Wang, X. Zhang, Peiqing Lv, H. Wang, and Y. Cheng, "Automatic Liver Segmentation Using EfficientNet and Attention-Based Residual U-Net in CT," Journal of Digital Imaging, vol. 35, no. 10, Jun. 2022.

[8] M. Sohaib Aslam et al., "Liver-Tumor Detection Using CNN ResUNet," Computers, Materials & Continua, vol. 67, no. 2, pp. 1899–1914, 2021.

[9] X. Wu et al., "Diagnosis Assistant for Liver Cancer Utilizing a Large Language Model with Three Types of Knowledge," arXiv.org, 2024.

[10] K. C. Kaluva, M. Khened, A. Kori, and G. Krishnamurthi, "2D-Densely Connected Convolution Neural Networks for automatic Liver and Tumor Segmentation," arXiv.org, 2018.

[11] Y. Masutani, K. Uozumi, M. Akahane, and K. Ohtomo, "Liver CT image processing: A short introduction of the technical elements," European Journal of Radiology, vol. 58, no. 2, pp. 246–251, May 2006.